ASA-EmulatR V-4

Table of Contents

1 Introduction

Moniker: INTRO

1.1 Configuring

Moniker: INTRO-CONFIG

Configuration precedence (lowest to highest): compiled-in struct defaults < EmulatrV4.ini < command-line qualifiers. The committed ini drives a no-flag launch; a CLI qualifier overrides the ini for a single run. IniLoader populates EmulatorSettings, which is the single source of truth threaded to every consumer.

EmulatR - Configuration

Field	ini Key	default(ini)	Example	Qualifier - Environment Override
model	[System]./model	DS10	ds10	
cpuCount	[System]./cpuCount	1		
port	[SRMConsole]./port	10023		EMULATR_CONSOLE_PORT
puttyPath	[SRMConsole]./puttyPath	putty.exe	C:\Program Files\PuTTY\putty.exe	
firmwareImage	[ROM]./firmwareImage	ds10_v7_3.exe		--firmware
firmwareSha256	[ROM]./firmwareSha256	*(empty - integrity check skipped)*		

Command-line qualifiers override the ini for a single run. Live today: --firmware (firmware image path), --mem (memory size). Planned (locked, landing with the single-source-of-truth work): --console-port, --platform-config, --flash-path; the EMULATR_CONSOLE_PORT / EMULATR_FLASH_ROM / EMULATR_PLATFORM_CONFIG environment variables remain as documented aliases.
Debug-only environment variables (EMULATR_IDE_TRACE, EMULATR_IIC_TRACE, EMULATR_FLASH_TRACE, EMULATR_NO_PUTTY) are diagnostic toggles, not configuration: they have no ini key, no CLI qualifier, and no precedence interaction.

EmulatR - Bus Configuration

Bus configuration is loaded for the configured model-platform, e.g., DS10 - bus configuration { build/run directory: `ds10_platform.json`}

Platform	Bus Configuration
DS10	ds10_platform.json
ES40	es40_platform.json
ES45	es45_platform.json

The platform file is selected from model: lower(model) + "_platform.json" (e.g. DS10 -> ds10_platform.json). An explicit override wins, in order: EMULATR_PLATFORM_CONFIG environment variable, then a platformConfigPath ini key, then the derived name. A missing platform file is a hard error -- there is no silent fallback.

1.1.1 Console Support

Moniker: INTRO-CONSOLE

Overview

EmulatR exposes a console server over a TCP socket.
The console is implemented as a RAW TCP stream and may be accessed by terminal applications such as:

- PuTTY
- Plink
- Tera Term
- MobaXterm
- netcat (nc)
- telnet (when supported)
- custom terminal software

Default Configuration

Example:

```
Host: localhost
Port: 10023
Protocol: RAW TCP
```

If EmulatR is running on another system, replace "localhost" with the hostname or IP address of the emulator host.
Example:

```
Host: 192.168.1.50
Port: 10023
```

Plink

Connect to a local EmulatR instance:

```
plink -raw localhost -P 10023
```

Connect to a remote EmulatR instance:

```
plink -raw 192.168.1.50 -P 10023
```

Enable verbose diagnostics:

```
plink -v -raw localhost -P 10023
```

Useful Plink Arguments

```
-raw
      Connect using a raw TCP stream.

-P <port>
      Specify TCP port.

-v
      Enable verbose diagnostics.

-log <file>
      Save session log.
```

Example:

```
plink -raw localhost -P 10023 -log emulator.log
```

PuTTY

1. Launch PuTTY.
2. Enter:

```
Host Name:
      localhost
Port:
      10023
```

3. Select:

```
Connection Type:
```

```
Raw
```

4. Press Open.

Remote Example:

```
Host Name:
      192.168.1.50
Port:
      10023
Connection Type:
      Raw
```

Tera Term

1. Launch Tera Term.
2. Select:

```
TCP/IP
```

3. Enter:

```
Host:
      localhost
TCP Port:
      10023
```

4. Connect.

MobaXterm

1. Session
2. Network Session
3. Configure:

```
Remote Host:
      localhost
Port:
      10023
Protocol:
      TCP
```

4. Connect.

Netcat (Linux/macOS)

Connect:

```
nc localhost 10023
```

Remote:

```
nc 192.168.1.50 10023
```

Useful for scripting and automated testing.

Telnet

Some telnet clients may connect successfully:

```
telnet localhost 10023
```

However, because EmulatR uses a raw socket and does not perform Telnet option negotiation, a raw-capable client is preferred.
Recommended clients:

```
PuTTY (Raw)
Plink (-raw)
Tera Term (TCP/IP)
MobaXterm (TCP)
```

Connection Verification

After connecting, the user should observe console output similar to:

```
EmulatR Console Ready
```

or

```
>>>
```

or

```
SRM>
```

depending on the current emulator state.

Troubleshooting

Connection Refused

```
Verify EmulatR is running.
Verify the console server is enabled.
Verify the configured port number.
```

Connection Timeout

```
Verify firewall configuration.
Verify the correct host address.
Verify the console listener is active.
```

No Console Output

```
Verify emulator boot has completed.
Verify console output is enabled.
Verify the correct port is being used.
```

Recommended Configuration

For emulator development and debugging:

```
Client:
      Plink

Command:

      plink -v -raw localhost -P 10023
```

This configuration provides:

```
- Raw TCP access
- Diagnostic connection logging
- Scriptability
- Low overhead
- Easy automation
```

Example Startup Workflow

1. Start EmulatR.
2. Verify console listener:

```
localhost:10023
```

3. Open terminal:

```
plink -raw localhost -P 10023
```

4. Interact with the emulator console.
5. Open additional sessions as desired for monitoring and debugging.

1.2 Executing

Moniker: INTRO-EXEC

Bash Command Line	Description
`EMULATR_NO_AUTOLOAD`=1 `EMULATR_CONSOLE_PORT`=10024 \ `EMULATR_IDE_TRACE`=1 `EMULATR_IIC_TRACE`=1 `EMULATR_FLASH_TRACE`=1 \ ./out/build/release/Emulatr.exe \ --firmware firmware/ds10_v7_3.exe --autosnapshot off 2> run_ide_release.log	PuTTY will auto-launch on 10024 (it follows the port). Qualifier command line-switch `--firmware firmware/ds10_v7_3.exe` explicit rather than relying on the ini fallback, because the ini value is the bare ds10_v7_3.exe (no firmware/ prefix) and would resolve against CWD.

`EMULATR_NO_AUTOLOAD=1`
`EMULATR_CONSOLE_PORT=1002`
`EMULATR_CONSOLE_MIRROR=1`
`EMULATR_IDE_TRACE=1`
`EMULATR_IIC_TRACE=1`
`EMULATR_FLASH_TRACE=1 ./o`
	Run the concurrent traced + mirrored boot (console on 10024 so it coexists with your other instance):

`# 2. cold boot, autosnaps`
`#    no cycle cap = indef`
`nohup env EMULATR_NO_AUTO`
`  ./out/build/relwithdebi`
`  > run_overnight_$(date`
`echo "PID $! — leave this`
	# 0. make sure nothing else is bound to the console port
	netstat -ano | grep 10023 || echo "port free"
	
	# 1. archive the stale (pre-SuperIO) snapshots instead of deleting
	mkdir -p snapshots/_pre_superio_archive
	mv snapshots/*.axpsnap snapshots/_pre_superio_archive/ 2>/dev/null; ls snapshots/
	
	# 2. cold boot, autosnapshot ON (default), marker ARMED, console mirror on,
	# no cycle cap = indefinite. nohup so it survives logoff.
	nohup env EMULATR_NO_AUTOLOAD=1 EMULATR_CONSOLE_SNAPSHOT=1 EMULATR_CONSOLE_MIRROR=1 EMULATR_CONSOLE_PORT=10023 \
	  ./out/build/relwithdebinfo/Emulatr.exe --firmware firmware/ds10_v7_3.exe \

Bash Command Line	Description
	> run_overnight_$(date +%Y%m%d_%H%M%S).log 2>&1 & echo "PID $! — leave this session alive overnight"

1.2.1 Executing Two Instances

Moniker: INTRO-EXEC2

One caveat for running two instances from the same directory: they share ds10_flash.rom and snapshots/, and concurrent NVRAM writes can corrupt the flash. If this run will write NVRAM (any set/update at >>>), isolate it first:

```
cp ds10_flash.rom ds10_flash_release.rom
# then prepend:  EMULATR_FLASH_ROM=ds10_flash_release.rom
```

After it's up, confirm and harvest in one shot:

```
grep -E 'SRM Console|IDE-TRACE|IIC-TXN|FLASH' run_ide_release.log | head -
100
```

You want to see Listening on TCP port 10024. Paste that grep output and I'll classify #1 (dqa0), #4 (IIC), and #11 (flash) — and the SSOT changes get a live confirmation too: the boot log should show the port coming from the ini path now, and a no---firmware launch would pick up the DS10 image.

1.3 Example - Performance

Moniker: INTRO-PERF

```
Enter
ASA EmulatR -- Alpha AXP (EV6 / 21264) Emulator
Alpha Emulator Console V4.0-0
(c) 2026 Timothy Peer / eNVy Systems, Inc.

*** keyboard not plugged in...
Flash ROM writes are disabled
64 Meg of system memory
probing hose 0, PCI
probing PCI-to-ISA bridge, bus 1
bus 0, slot 5 -- dqa -- Cypress 82C693 IDE
initializing GCT/FRU at 3f32000

Checking dqa0.0.0.105.0 for the option firmware files. . .
dqa0.0.0.105.0 has no media present or is disabled via the RUN/STOP switch
Checking dva0.0.0.0.0 for the option firmware files. . .

Option firmware files were not found on CD or floppy.
If you want to load the options firmware,
please enter the device on which the files are located(ewa0),
```

```
or just hit <return> to proceed with a standard console update:


                ***** Loadable Firmware Update Utility *****
-------------------------------------------------------------------------------
 Function    Description
-------------------------------------------------------------------------------
 Display     Displays the system's configuration table.
 Exit        Done exit LFU (reset).
 List        Lists the device, revision, firmware name, and update
revision.
 Readme      Lists important release information.
 Update      Replaces current firmware with loadable data image.
 Verify      Compares loadable and hardware images.
 ? or Help   Scrolls this function table.
-------------------------------------------------------------------------------

UPD> update srm

Confirm update on:
srm              [Y/(N)]y

WARNING: updates may take several minutes to complete for each device.

                          DO NOT ABORT!

srm             Updating to 7.3-1...  Verifying 7.3-1...  PASSED.

UPD> exit

 Initializing....
Testing the System
Testing the Disks (read only)
exer: No such command

snapshot Console V7.3-2, Feb 27 2007 13:25:53

Halt Button is IN, AUTO_ACTION ignored

>>>show config
                        snapshot

SRM Console:    V7.3-2
PALcode:        OpenVMS PALcode V1.98-83, Tru64 UNIX PALcode V1.92-73

Processors
CPU 0           Alpha EV6 pass 1 266 MHz
                SROM Revision:   P▒
                Bcache is disabled

Core Logic
Cchip           DECchip 21272-CA Rev 2
Dchip           DECchip 21272-DA Rev 1
```

```
Pchip 0          DECchip 21272-EA Rev 1

TIG              Rev 7.31
Arbiter          Rev 7.31 (0xff)

MEMORY

Array #       Size      Base Addr
-------    ----------  ---------
   0          64 MB    000000000

Total Bad Pages = 0
Total Good Memory = 64 MBytes


PCI Hose 00
     Bus 00  Slot 05/0: Cypress 82C693
                                                         Bridge to Bus 1,
ISA
     Bus 00  Slot 05/1: Cypress 82C693 IDE
                                   dqa.0.0.105.0
                                   dqa0.0.0.105.0         EMULATR VIRTUAL
CDRO


ISA
Slot    Device  Name            Type         Enabled  BaseAddr  IRQ     DMA
0
        0       MOUSE           Embedded        Yes      60      12

        1       KBD             Embedded        Yes      60      1

        2       COM1            Embedded        Yes      3f8     4

        3       COM2            Embedded        Yes      2f8     3

        4       LPT1            Embedded        Yes      3bc     7

        5       FLOPPY          Embedded        Yes      3f0     6       2


>>>show devices
>>>show device
dqa0.0.0.105.0             DQA0          EMULATR VIRTUAL CDROM
dva0.0.0.0.0               DVA0
>>>show memory_test
memory_test             full
>>>show full_powerup_diags
full_powerup_diags      ON
>>>set oem_string snapshot
>>>set memory_test none
>>>set full_powerup_diags off
>>>show memory_test
```

```
memory_test              none
>>>show full_powerup_diags
full_powerup_diags       OFF
>>>
```

1.4 What is EmulatR

Moniker: INTRO-WHATIS

EmulatR is a software emulator of the Digital (DEC) Alpha AXP family. It models the 21264 / 21264A (EV6 / EV67) processor and the 21272 “Tsunami / Typhoon” core-logic chipset, targeting AlphaServer DS10 / ES40 / ES45-class machines.
Its goal is faithful, cycle-aware emulation — complete enough to load and execute original, unmodified DEC SRM console firmware and reach the interactive >>> prompt. The core is written in modern C++ (C++20) and is kept free of any GUI framework dependency; the console subsystem layers on top of it.

At a glance:

- CPU: Alpha 21264A (EV6 / EV67), OpenVMS and Tru64 UNIX PALcode.
- Core logic: 21272 Tsunami / Typhoon — Cchip (21272-CA), Dchip (21272-DA), Pchip (21272-EA).
- Southbridge: Cypress CY82C693 (PCI-to-ISA bridge, IDE/ATAPI, SuperIO: FDC, COM1/COM2, LPT1).
- Firmware: runs a real SRM image (for example, DS10 SRM V7.3, ds10_v7_3.exe).
- Console: RAW TCP server, reached with any terminal client (PuTTY, plink, etc.).

1.5 Supported Systems

Moniker: INTRO-SUPPORTED

EmulatR targets the Tsunami / Typhoon AlphaServer line. The emulated hardware reported by the SRM console (show config) reflects a DS10-class machine:

- Processor: Alpha EV6 pass 1, 266 MHz, Bcache disabled.
- Core logic: DECchip 21272-CA (Cchip) Rev 2, 21272-DA (Dchip) Rev 1, 21272-EA (Pchip 0) Rev 1; TIG / Arbiter Rev 7.31.
- Memory: 64 MB default array, base 0x000000000.
- I/O: PCI Hose 00 with the Cypress 82C693 bridge/IDE at bus 0, slot 5; ISA devices MOUSE, KBD, COM1, COM2, LPT1, FLOPPY.

Platform selection is driven by a per-model bus-configuration manifest (see Configuring). DS10 is the actively developed and validated platform; ES40 and ES45 manifests are scaffolded (see Current Platform Support).

1.6 Current Platform Support

Moniker: INTRO-PLATFORM

A platform manifest is selected from the model name as lower(model) + "_platform.json" (for example, DS10 → ds10_platform.json). The table below summarizes current status. "Validated" means the platform has been driven to the SRM >>> prompt in EmulatR.

Platform	Manifest	CPU / Chipset	Status
DS10	ds10_platform.json	EV6 / 21272	Active — boots to >>>
ES40	es40_platform.json	EV6 / 21272	Manifest present — not yet validated
ES45	es45_platform.json	EV67 / 21272	Manifest present — not yet validated

A missing platform file is a hard error — there is no silent fallback. The selection order is: EMULATR_PLATFORM_CONFIG, then a platformConfigPath ini key, then the derived name.

1.7 Feature Matrix

Moniker: INTRO-FEATUREMATRIX

Subsystem status as of June 2026. This reflects the DS10 target; status is expected to evolve and should be reconfirmed against the current build.

Subsystem	Status	Notes
CPU pipeline (EV6) + TLB	Working	128 ITB + 128 DTB; PALcode relocation in place.
SRM firmware boot → >>>	Working	Reaches the interactive console prompt.
TCP console server	Working	RAW TCP; PuTTY auto-launch.
Snapshot save / restore	Working	.axpsnap; autoload-newest, marker capture.
SuperIO (FDC / COM / LPT)	Working	CY82C693 SuperIO word-fold confirmed.
IDE / ATAPI virtual CD-ROM	In progress	dqa0 enumeration gap under investigation.
IIC (PCF8584)	Under investigation	S1 returns 0x01 where firmware expects 0x00.
Flash / NVRAM persistence	Under investigation	"Flash ROM writes are disabled" — see Known Limitations.

Subsystem	Status	Notes
boot / OS load (auto_action BOOT)	Planned	HALT is the only fully implemented action.
SCSI	Planned	[TODO: confirm — no SCSI HBA currently modeled.]

2 Installation

Moniker: INSTALL

This chapter covers obtaining the tools, building EmulatR from source, supplying an SRM firmware image, and the first launch. Build specifics below are marked where they still need to be confirmed against the current project files.

2.1 Building from Source

Moniker: INSTALL-BUILD

EmulatR is a CMake project built with MSVC (C++20). The build tree produces Emulatr.exe under out/build/<config>/, with the two configurations in active use being release and relwithdebinfo (the latter is the day-to-day debug/profiling build).

General shape of a build (exact preset and generator names to be confirmed):

```
# configure
cmake --preset <preset>        # [TODO: confirm preset name]
# build (RelWithDebInfo is the standard working build)
cmake --build --preset <preset> --config RelWithDebInfo
# resulting binary
./out/build/relwithdebinfo/Emulatr.exe
```

[TODO: confirm and document the exact configure/build commands, CMake generator, and any dependency provisioning (e.g. Qt and spdlog acquisition) for a clean checkout.]

Note: the diagnostic build flag EMULATR_MEMDIAG is a CMake option, not a runtime variable, and must stay OFF for normal use (its per-instruction logging imposes roughly a 900x slowdown).

2.2 Required Tools - Frameworks

Moniker: INSTALL-TOOLS

Tooling and libraries used by EmulatR. Exact version requirements are to be confirmed.

- Compiler: MSVC with C++20 support. [TODO: confirm minimum toolset version.]
- Build system: CMake. [TODO: confirm minimum version.]
- Qt: used by the console / UI subsystem only — the emulator core is deliberately Qt-free. [TODO: confirm version.]
- spdlog: logging infrastructure (dedicated `EMULATR_CONSOLE_TRACE` channel). [TODO: confirm version.]
- A terminal client for the console: PuTTY / plink (auto-launched), or any RAW-TCP client.
- Optional: Ghidra, for SRM firmware reverse-engineering work.

2.3 Firmware Requirements

Moniker: INSTALL-FIRMWARE

EmulatR requires an SRM console firmware image to execute. The DS10 reference image is an SRM V7.3 build (ds10_v7_3.exe).

The image is selected, in precedence order:
- --firmware <path> on the command line (overrides the ini for a single run); or
- the [ROM]/firmwareImage ini key (default: ds10_v7_3.exe).

Path gotcha: the ini value is the bare file name (no firmware/ prefix) and resolves against the current working directory. Launch from the Emulatr root, or pass an explicit --firmware firmware/ds10_v7_3.exe.
Optional integrity check: set [ROM]/firmwareSha256 to verify the image at load. When empty, the integrity check is skipped.
[TODO: document firmware provenance, licensing, and the set of supported SRM versions.]

2.4 First Startup

Moniker: INSTALL-FIRSTSTART
A first run, launched from the Emulatr root so relative paths resolve correctly:

```
cd /d/EmulatR/EmulatRAppUniV4/Emulatr
./out/build/relwithdebinfo/Emulatr.exe --firmware firmware/ds10_v7_3.exe
```

On launch the TCP console server comes up (default port `10023`) and, unless `EMULATR_NO_PUTTY` is set, a PuTTY session is auto-launched against it. The firmware then runs through its power-up sequence to the SRM `>>>` prompt. For the step-by-step walkthrough, see Quick Start; to stop cleanly, see Clean Shutdown.

3 Quick Start

Moniker: QS

The fastest path from a built binary to an interactive SRM prompt, and back down cleanly. Each step has its own topic below.

3.1 Launching EmulatR

Moniker: QS-LAUNCH

Launch from the Emulatr root. A genuine cold boot (no snapshot restore), quiet, is:

```
cd /d/EmulatR/EmulatRAppUniV4/Emulatr
EMULATR_NO_AUTOLOAD=1 ./out/build/relwithdebinfo/Emulatr.exe \
  --firmware firmware/ds10_v7_3.exe --autosnapshot off  2> run.log
```

Omit `EMULATR_NO_AUTOLOAD` to restore the newest snapshot instead of cold-booting. Diagnostics are written to the redirected stderr (run.log). See the appendix Environment Variables reference for the full set of launch gates.

3.2 Connecting to the Console

Moniker: QS-CONNECT

The emulator exposes a RAW TCP console (default localhost:`10023`). Unless `EMULATR_NO_PUTTY` is set, PuTTY is launched for you. To attach a client manually:

```
plink -raw localhost -P 10023
```

Full client coverage (PuTTY, plink, Tera Term, MobaXterm, netcat, telnet) is in the Console Connectivity chapter.

3.3 Loading Firmware

Moniker: QS-FIRMWARE

Pass the SRM image explicitly so the path resolves regardless of the ini default:

```
--firmware firmware/ds10_v7_3.exe
```

Relying on the ini fallback resolves the bare name ds10_v7_3.exe against the current directory, so an explicit --firmware is the reliable form. See Firmware Requirements for selection precedence and the optional SHA-256 integrity check.

3.4 Reaching the SRM Prompt

Moniker: QS-SRMPROMPT

After launch the firmware proceeds through power-up to the console. The visible sequence is:

- Banner and memory sizing (“`64 Meg of system memory`”).

- PCI / ISA probe; GCT/FRU initialization at `3f32000`.
- Option-firmware scan of the CD / floppy devices.
- The Loadable Firmware Update (LFU) utility — the `UPD>` prompt. From here, exit returns to the console, or update srm refreshes firmware.
- System / disk testing, then the interactive `>>>` prompt.

A full annotated transcript is in Example - Performance. By default the console halts at `>>>` (`auto_action = HALT`).

3.5 Clean Shutdown

Moniker: QS-SHUTDOWN
There are two graceful paths and one hard kill.

1. Stop sentinel (flushes NVRAM via ~Machine):

```
touch /d/EmulatR/EmulatRAppUniV4/Emulatr/EMULATR_STOP
```

The run loop polls the sentinel even mid-stall, exits within a moment, and writes the flash cleanly. The resolved sentinel path is logged at startup (grep graceful-stop run.log).

2. CALL_PAL HALT at the prompt:

Halting at >>> exits through ~Machine with a clean flush — no sentinel touch needed.

3. Hard kill (instant, no flush):

```
taskkill //F //IM Emulatr.exe
taskkill //F //IM PuTTY.exe
```

See also: Kill the EmulatR (appendix).

4 Console Connectivity

Moniker: CON

The console is a RAW TCP stream; any raw-capable terminal client can attach. The detailed reference (default configuration, recommended setup, connection verification, troubleshooting) lives under Configuring → Console Support; the per-client topics below are the quick reference.

4.1 TCP Console Overview

Moniker: CON-OVERVIEW

EmulatR serves the console over a RAW TCP socket — no Telnet option negotiation. Defaults:

```
Host: localhost
Port: 10023
Protocol: RAW TCP
```

Override the port with `EMULATR_CONSOLE_PORT` (PuTTY auto-launch follows the port). Suppress the PuTTY 22 auto-launch with `EMULATR_NO_PUTTY`; the listen socket still comes up for your own client. For a remote host, replace localhost with its hostname or IP.

Port `10023` is only the default — the console listener can be bound to any available TCP port. Override it at launch with the `EMULATR_CONSOLE_PORT` environment variable (for example, `EMULATR_CONSOLE_PORT`=10024); the auto-launched PuTTY session follows whatever port you set. This is what makes it practical to run more than one instance at once: give each its own port (10023, 10024, 10025, ...) so their consoles don't collide. Choose a free port above the privileged range (1024 and up) that nothing else on the host is already using — `netstat -ano | grep <port>` confirms a port is free before you bind it. When connecting, point your client at the same port the instance was started with.

4.2 PuTTY

Moniker: CON-PUTTY

1. Launch PuTTY.
2. Host Name: localhost   Port: `10023`
3. Connection Type: Raw
4. Press Open.

For a remote instance, substitute the host IP (for example, `192.168.1.50`) and keep Connection Type = Raw.

4.3 Plink

Moniker: CON-PLINK

Connect to a local instance:

```
plink -raw localhost -P 10023
```

Remote instance:

```
plink -raw 192.168.1.50 -P 10023
```

Verbose diagnostics and a session log:

```
plink -v -raw localhost -P 10023 -log emulator.log
```

Key arguments: -raw (raw TCP), -P <port>, -v (verbose), -log <file>.

4.4 TeraTerm

Moniker: CON-TERATERM

1. Launch Tera Term.
2. Select TCP/IP.
3. Host: localhost   TCP Port: 10023
4. Connect.

4.5 MobaXterm

Moniker: CON-MOBAXTERM

1. Session → Network Session.
2. Remote Host: localhost   Port: `10023`   Protocol: TCP
3. Connect.

4.6 Netcat

Moniker: CON-NETCAT

Local:

```
nc localhost 10023
```

Remote:

```
nc 192.168.1.50 10023
```

Useful for scripting and automated testing. (On telnet: most clients connect, but because the socket is raw with no option negotiation, a raw-capable client is preferred.)

4.7 Troubleshooting

Moniker: CON-TROUBLE

Connection refused:

- Verify EmulatR is running and the console server is enabled.
- Verify the configured port number (default 10023; `EMULATR_CONSOLE_PORT` to change).

Connection timeout:

- Verify firewall configuration and the correct host address.
- Verify the console listener is active (look for “Listening on TCP port ...” in the log).

No console output:

- Verify emulator boot has completed and console output is enabled.
- Verify the correct port is being used.

5 SRM Console

Moniker: SRM

SRM is the Alpha system console firmware. It presents the >>> prompt, stores configuration in NVRAM as environment variables, and (on real hardware) bootstraps the operating system. On EmulatR the console runs to >>> and halts; OS bootstrap is planned.

5.1 SRM Overview

Moniker: SRM-OVERVIEW

At >>> you interact with the console using show (display a variable), set (change a variable), and device/diagnostic commands. Settings persist to the flash/NVRAM image. The default power-up action is HALT, so the console waits at >>> for an operator command.
The reported console version is V7.3-2 (Feb 27 2007); PALcode is OpenVMS V1.98-83 / Tru64 UNIX V1.92-73.

5.2 Common Commands

Moniker: SRM-COMMON

- `show <var>` — display the stored and active value of a console variable.
- `set <var> <value>` — change a variable (persisted to NVRAM).
- `show config` — system configuration table (CPU, core logic, memory, PCI/ISA).
- `show device` — enumerated storage/IO devices.
- `show memory` — memory array layout.
- `boot` — bootstrap the OS (planned on EmulatR; HALT is the implemented action).
- `help` / ? — console help.

5.3 show config

Moniker: SRM-SHOWCONFIG

show config prints the full configuration table: the SRM and PALcode versions, the processor (Alpha EV6 pass 1, 266 MHz), core logic (Cchip 21272-CA, Dchip 21272-DA, Pchip 21272-EA, TIG/Arbiter Rev 7.31), the memory array, the PCI hose (Cypress 82C693 bridge and IDE at bus 0 slot 5), and the ISA device list (MOUSE, KBD, COM1, COM2, LPT1, FLOPPY with their base addresses and IRQs). A full sample is in Example - Performance.

5.4 show device

Moniker: SRM-SHOWDEV

show device lists enumerated devices. On the DS10 target:

```
dqa0.0.0.105.0    DQA0    EMULATR VIRTUAL CDROM
dva0.0.0.0.0      DVA0
```

dqa0 is the virtual ATAPI CD-ROM on the CY82C693 IDE channel; dva0 is the floppy device. (The dqa0 enumeration path is under active work — see Storage Devices and Known Limitations.)

5.5 show memory

Moniker: SRM-SHOWMEM

The memory array is reported in the configuration table:

```
Array #      Size        Base Addr
   0         64 MB       000000000
Total Good Memory = 64 MBytes,  Total Bad Pages = 0
```

Default system memory is 64 MB; the size can be set at launch with `--mem`.

5.6 boot

Moniker: SRM-BOOT

On real SRM, boot bootstraps the operating system using bootdef_dev and boot_osflags. On EmulatR, OS bootstrap is not yet implemented — the console reaches `>>>` and halts. The related auto_action variable supports HALT (default and only fully implemented action); BOOT and RESTART are planned.

5.7 set

Moniker: SRM-SET

set changes a console variable and persists it to the flash/NVRAM image; show reads it back. Example session:

```
>>>set oem_string snapshot
>>>set memory_test none
>>>set full_powerup_diags off
>>>show memory_test
memory_test             none
```

Setting memory_test none and full_powerup_diags off shortens the power-up self-test — useful for faster iteration (see Slow Boot).

5.8 SRM Environment Variables

Moniker: SRM-ENVVARS

The complete, annotated list of SRM console variables (auto_action, boot_*, com1_*/com2_*, the d_* diagnostics group, the N1–N16 NVRAM slots, wwid0–wwid7, and so on) is maintained in the appendix:

- Appendix → DS10 → SRM Commands >>> Prompt (moniker APP-DS10-SRMCMDS).

These SRM variables are distinct from the emulator's own EMULATR_* launch environment variables, which are documented under Environment Variables - Operation Reference (APP-DS10-ENVVARS).

6 Storage Devices

Moniker: STOR

Storage on the DS10 target is presented through the Cypress CY82C693 southbridge IDE/ATAPI channels. SCSI and general virtual-disk support are noted below where status is still to be confirmed.

6.1 IDE Devices

Moniker: STOR-IDE

The CY82C693 provides the IDE controller, exposed on PCI bus 0, slot 5, function 1 (“Cypress 82C693 IDE” in show config). The primary channel is the dqa device path. IDE register/port behaviour (0x1F0–0x1F7 and PCI config) and absent-slave probing are part of the current device-enumeration work.

6.2 ATAPI Devices

Moniker: STOR-ATAPI

The CD-ROM is presented as an ATAPI device on the IDE channel — dqa0.0.0.105.0, “EMULATR VIRTUAL CDROM”. ATAPI/IDE scaffolding for the CY82C693 is in active development; the dqa0 enumeration gap (firmware does not consistently see the device during the option-firmware scan) is a known open item (see Missing Devices and Known Limitations).

6.3 SCSI Devices

Moniker: STOR-SCSI

The SRM console exposes SCSI-related variables (scsi_poll, scsi_reset), but EmulatR does not currently model a SCSI host adapter. [TODO: confirm — SCSI is not implemented; treat as planned.]

6.4 Virtual CDROM

Moniker: STOR-CDROM

`dqa0` (“EMULATR VIRTUAL CDROM”) is the virtual optical device the firmware probes for option firmware during power-up. When no media is present it reports “no media present or is disabled via the RUN/STOP switch”. [TODO: document how an ISO / media image is attached to dqa0.]

6.5 Virtual Disk Images

Moniker: STOR-DISK

[TODO: confirm and document virtual hard-disk image support — supported container format(s), how an image is attached, and the SRM device path it appears under. Not yet documented.]

7 Snapshot Support

Moniker: SNAP

EmulatR can capture and restore full machine state to .axpsnap files, so a long cold boot need only be paid once. Snapshot behaviour is controlled by the EMULATR_* environment variables and the matching --snapshot / --autosnapshot CLI flags.

7.1 Automatic Snapshots

Moniker: SNAP-AUTO

Periodic auto_*.axpsnap saves are on by default. Disable them for clean cold-boot / profiling runs:

```
--autosnapshot off            # or  EMULATR_AUTOSNAP=off
```

On startup the newest snapshot is auto-loaded unless you force a cold boot:

```
--no-autoload                 # or  EMULATR_NO_AUTOLOAD=1
```

Disabling autosnapshot on mint cold-boot runs avoids the per-save disk cost.

7.2 Creating Snapshots

Moniker: SNAP-CREATE

Two capture mechanisms:

- Console marker: arm with EMULATR_CONSOLE_SNAPSHOT=1; when a console input line matches the marker (default: “`set oem_string snapshot`”), a ./snapshots/predig_oemsnap_cyc<N>.axpsnap is written. The marker text can be changed with `EMULATR_SNAPSHOT_MARKER` (leading/trailing whitespace trimmed; internal whitespace is NOT normalized).
- PC-triggered: --snapshot-on-pc <pc>[,...], optionally named with `--snapshot-name-tag <s>`.

7.3 Restoring Snapshots

Moniker: SNAP-RESTORE

By default, startup restores the newest snapshot automatically (autoload-newest). To ignore snapshots and force a genuine cold boot, launch with `--no-autoload` (`EMULATR_NO_AUTOLOAD=1`). Snapshot files are resolved in the CWD-relative snapshots/ directory, so launch from the Emulatr root for the canonical set.

7.4 Snapshot Pruning

Moniker: SNAP-PRUNE

Pruning of old snapshots is off by default — without the gate, pruneOldSnapshots is a no-op and nothing is evicted. Enable it with:

```
EMULATR_SNAPSHOT_PRUNE=1
```

8 Troubleshooting

Moniker: TRBL

Common failure modes and where to look. Several long pauses during boot are understood, time-based firmware loops rather than emulator faults — see Firmware Hangs 29 and Slow Boot 29 .

8.1 EmulatR will not start

Moniker: TRBL-NOSTART

- Console port already in use: check with `netstat -ano | grep 10023` and free it or set `EMULATR_CONSOLE_PORT`.
- Firmware not found: the ini default is the bare name and resolves against the CWD — pass `--firmware ./firmware/ds10_v7_3.exe` or launch from the Emulatr root.
- Missing platform manifest: a missing <model>_platform.json is a hard error with no fallback.
- Two instances from one directory share ds10_flash.rom and snapshots/; concurrent NVRAM writes can corrupt the flash. Isolate with `EMULATR_FLASH_ROM=`<copy> (see Executing Two Instances).

8.2 Firmware Hangs

Moniker: TRBL-FWHANG

Some long pauses are time-based firmware loops, not emulator hangs:

- Option-firmware scan: after "`Checking dva0.0.0.0.0 for the option firmware files. . .`" the firmware enters a `krn$_sleep`-based wait (the `0x7bef0` software-tick loop). This is a timed sleep, not a device probe — a profiled gap of roughly 36 minutes was observed here - opened item.
- dqa0 "no media present" is followed by a stall (suspected timer loop) .
- `GCT/FRU initialization at 3f32000` is significantly delayed.

See Known Limitations for the current status of each.

8.3 Slow Boot

Moniker: TRBL-SLOWBOOT

Boot time is dominated by power-up diagnostics and several bounded timeout loops. To shorten iteration:

```
>>>set memory_test none
>>>set full_powerup_diags off
```

Bounded device-wait loops (approximate wall-clock): `dq_wait_on_busy` ~5000×100µs; `dq_wait_for_drq` ~5000×100–200µs; `dq_long_wait_on_busy` ~310000×100µs (~31 s).

The experimental `EMULATR_TICKWARP` gate fast-forwards known delay loops but alters timing fidelity.

8.4 Missing Devices

Moniker: TRBL-MISSDEV

If a device is absent from show device, check the IDE/ATAPI enumeration path. The dqa0 (virtual CD-ROM) enumeration gap is a known open item: the firmware does not always observe the device during the option-firmware scan. Tracing `0x1F0–0x1F7` plus the PCI config of bus0/dev5/func1 distinguishes a port-never-initialized case from a probe-protocol mismatch.

8.5 Console Connection Issues

Moniker: TRBL-CONN

For refused / timed-out / silent console connections, see the Console troubleshooting reference (Console Support, moniker CON-TROUBLE). In brief: confirm the listener is up ("`Listening on TCP port …`"), the port matches your client, and — since the socket is raw — the client is in RAW mode.

9 Appendix Master

Moniker: APP

Per-platform reference material and deep-dive engineering notes. The DS10 appendix is the most developed; DS20 / ES40 / ES45 are placeholders pending platform validation.

9.1 Appendix - DS10

Moniker: APP-DS10

Reference material specific to the DS10 platform: the emulator environment-variable operation reference, the SRM `>>>` variable catalogue, the SuperIO controller analysis, the engineering task matrix, boot-timeout bounds, and clean-shutdown procedure. Sub-topics each carry their own moniker (APP-DS10-*).

9.1.1 Environment Variables - Operation Reference

Moniker: APP-DS10-ENVVARS

Generated 2026-06-08 from the `std::getenv` call sites in EmulatRAppUniV4\Emulatr. ASCII-only.

This document lists the **runtime** environment variables the emulator reads at launch, plus the **compile-time** flags that look like env vars but are actually CMake build options. CLI flags that mirror an env var are cross-referenced.

How they are read (important gotchas)

- All values are read with `std::getenv`. Set them in the launching shell before starting the process. Inline bash assignment is the normal form:

```
EMULATR_CONSOLE_SNAPSHOT=1 ./out/build/relwithdebinfo/Emulatr.exe --firmware firmwar
```

- **Read-once gotcha.** The trace/diagnostic gates (`EMULATR_CONSOLE_SNAPSHOT`, `EMULATR_IIC_TRACE`, `EMULATR_FLASH_TRACE`, `EMULATR_PIC_TRACE`, `EMULATR_TICKWARP`, the RPCC probe knobs) cache their result in a function-local `static bool const` on first call. They are effectively read once at process start; exporting them mid-run has no effect.

- **Presence vs value.** Most gates test only *presence* (`getenv(...) != nullptr`), so VAR=0 still ENABLES them -- any value, including `0` or empty, counts as set. The exceptions that parse a value are called out below (`EMULATR_AUTOSNAP`, `EMULATR_SNAPSHOT_MARKER`, the path vars, and the RPCC numeric knobs).

- **Relative paths resolve against the launch CWD.** `EMULATR_STOP_FILE` default and the `snapshots/` dir are CWD-relative; launch from the Emulatr root for the canonical snapshot set.

Snapshot / autoload

Variable	Values	Default	Effect
`EMULATR_CONSOLE_SNAPSHOT`	presence	unset (off)	Arms the Option-A console-snapshot marker watch in the UART. When set, a console input line matching the marker writes `snapshots/predig_oemsnap_cyc<N>.axpsnap`. Read once at start.
`EMULATR_SNAPSHOT_MARKER`	string	`set oem_string snapshot`	The exact line that triggers the console snapshot. Leading/trailing whitespace is trimmed; **internal whitespace is NOT normalized** (type single spaces).
`EMULATR_AUTOSNAP`	`off` (else on)	on	Disables periodic `auto_*.axpsnap` saves. Mirrors `--autosnapshot off`. Use for cold-boot mint runs to avoid the per-save disk cliff.
`EMULATR_NO_AUTOLOAD`	presence	unset	Skip autoload-newest at startup; forces a genuine cold boot instead of restoring the newest snapshot. Mirrors `--no-autoload`.
`EMULATR_SNAPSHOT_PRUNE`	presence	unset (no prune)	Enables pruning of old snapshots. Without it, `pruneOldSnapshots` is

Variable	Values	Default	Effect
			a no-op and nothing is evicted.

Process / platform control

Variable	Values	Default	Effect
`EMULATR_STOP_FILE`	path	`EMULATR_STOP` (CWD-relative)	Path of the graceful-stop sentinel. `touch` it to stop the run cleanly so `~Machine` flushes flash NVRAM. The resolved absolute path is logged at startup (`graceful-stop sentinel = ...`). Polled ~every 1M steps; consumed when seen.
`EMULATR_NO_PUTTY`	presence	unset	Do not auto-launch PuTTY for the TCP console backend. The listen socket still comes up; attach your own client.
`EMULATR_FLASH_ROM`	path	`ds10_flash.rom`	Backing file for the flash/NVRAM image (persisted config: env vars, `update srm` heal).
`EMULATR_PLATFORM_CONFIG`	path	built-in DS10 manifest	Override the platform manifest (IIC device population, etc.).

Logging / diagnostics (runtime gates)

Variable	Values	Default	Effect
`EMULATR_CHIPSET_DIAG_OFF`	presence/non-empty	unset (diag on)	Silences the chipset CSR diagnostic stream. Note the

Variable	Values	Default	Effect
			inverted sense: setting it turns diagnostics OFF.
`EMULATR_TRACE_WINDOW`	presence	unset	Forces the DecListingSink retire-compact trace window even without the `TRACE_RETIRE_COMPACT` mask bit.
`EMULATR_RETIRE_TRACE_DIR`	dir path	build-tree default	Output directory for retire-compact / breakpoint / profiler trace files.

Device traces (per-subsystem, read once)

Variable	Values	Default	Effect
`EMULATR_IIC_TRACE`	presence	unset	Trace PCF8584 IIC bus transactions.
`EMULATR_FLASH_TRACE`	presence	unset	Trace FlashRom command/erase/write activity.
`EMULATR_PIC_TRACE`	presence	unset	Trace the 8259 PIC pair (IRQ assert/EOI).

Performance / experimental

Variable	Values	Default	Effect
`EMULATR_TICKWARP`	presence	unset	Enables tick-warp fast-forward probes at known delay-loop PCs (RSCC balloon at `0x7c304`, spin at `0x1c655c`). *Experimental; alters timing fidelity.*
`EMULATR_RPCC_LOG_AFTER`	uint (dec/hex)	`185000000`	Cycle count after which the RPCC probe begins logging.

Variable	Values	Default	Effect
EMULATR_RPCC_LOG_MAX	uint (dec/hex)	200000	Max RPCC probe log entries.
EMULATR_RPCC_LOG_FILE	path	D:\EmulatR\EmulatRAppUniV4\rpcc_probe.txt	RPCC probe output file.

Probe / debug gates

Variable	Values	Default	Effect
EMULATR_GCT_WATCH	presence	unset	Store-watch on the GCT/FRU config-tree region (PA 0x3f32000-0x3f33fff).
EMULATR_BREAK_ON_PA10	presence	unset	Conditional break on accesses to PA 0x10-0x17.
EMULATR_CHECKPOINTS	spec string	unset	Named PC checkpoints, e.g. shcreate:0x...,shentry:0x...,isatty:0x...,rwp:0x....
EMULATR_GATE	spec string	unset	PC/condition gate spec controlling trace windowing.

Compile-time flags (NOT runtime env vars)

These appear as EMULATR_ names but are CMake options / preprocessor defines baked in at build time. Setting them in the shell does nothing; they must be enabled in the build.

Flag	Where	Notes
EMULATR_MEMDIAG	CMakeLists.txt:263 (commented out)	Per-retired-instruction fprintf+fflush. ~900x slowdown; **keep OFF**. Disabled 2026-06-01.
EMULATR_DIAGNOSTIC_LOGGING	CMakeLists.txt:315 option	Centralized per-subsystem diagnostic logging; defined only in non-Release configs. Mirrored onto the test target.

Flag	Where	Notes
EMULATR_PA_DUMP	CMakeLists.txt (test mirror)	PA-access dump; build define.
EMULATR_LOG_TRACE/DEBUG/INFO/WARN/ERROR/CRITICAL	logging macros	spdlog level selection macros, not runtime env.

CLI flags that mirror env vars

CLI	Env equivalent
--autosnapshot off	EMULATR_AUTOSNAP=off
--no-autoload	EMULATR_NO_AUTOLOAD=1
--max-cycles <N>	(no env; default unlimited = ~uint64_t{0})
--snapshot-on-pc <pc>[,...]	(no env)
--snapshot-name-tag <s>	(no env)

Operational recipes

Reach >>> and capture diagnostics, interactive (PuTTY console live):

```
cd /d/EmulatR/EmulatRAppUniV4/Emulatr
EMULATR_CONSOLE_SNAPSHOT=1 ./out/build/relwithdebinfo/Emulatr.exe \
  --firmware firmware/ds10_v7_3.exe --autosnapshot off  2> run.log
```

Same, fully detached (interaction is via the PuTTY/TCP console, not the shell):

```
cd /d/EmulatR/EmulatRAppUniV4/Emulatr
EMULATR_CONSOLE_SNAPSHOT=1 nohup ./out/build/relwithdebinfo/Emulatr.exe \
  --firmware firmware/ds10_v7_3.exe --autosnapshot off  > run.log 2>&1 &
disown; echo "pid=$!"
```

Stop a running instance gracefully (flushes flash NVRAM):

```
grep graceful-stop run.log          # find the exact sentinel path it
watches
touch /d/EmulatR/EmulatRAppUniV4/Emulatr/EMULATR_STOP
```

Genuine cold boot (no snapshot restore), quiet, for profiling:

```
EMULATR_NO_AUTOLOAD=1 ./out/build/relwithdebinfo/Emulatr.exe \
  --firmware firmware/ds10_v7_3.exe --autosnapshot off  2> /dev/null
```

Enable a single device trace (IIC bus, e.g. for the FRU/EEPROM path):

```
EMULATR_IIC_TRACE=1 ./out/build/relwithdebinfo/Emulatr.exe \
  --firmware firmware/ds10_v7_3.exe  2> iic.log
```

Maintenance: regenerate by grepping getenv across EmulatRAppUniV4\Emulatr and diffing against the tables above. New runtime gates should be added here and noted in their commit.

9.1.2 Kill the EmulatR

Moniker: APP-DS10-KILL

On canceling the running Emulatr.exe — two ways:

To cleanly stop the execution of the EmulatR.exe

[Bash] touch EMULATR_STOP

1. Clean stop (flushes NVRAM via ~Machine):

bash $touch /d/emulatr/emulatrappuniv4/emulatr/EMULATR_STOP

The run loop polls that sentinel even mid-stall, so it'll exit within a moment and write the flash cleanly.

2. Hard kill (instant, no flush):

bash taskkill //F //IM Emulatr.exe

bash taskkill //F //IM PuTTY.exe

9.1.3 Known Limitations

Moniker: APP-DS10-KNOWNLIM

Open items as of June 2026:

- Flash ROM writes are disabled — NVRAM persistence is under investigation.
- dqa0 (virtual CD-ROM) enumeration gap during the option-firmware scan.
- IIC (PCF8584) S1 returns `0x01` where firmware expects `0x00` at completion — benign-vs-blocking classification pending.
- Option-firmware scan enters a long `krn$_sleep` timed loop (0x7bef0); large boot-time gap.
- GCT/FRU initialization at `3f32000` is significantly delayed.
- auto_action BOOT / RESTART not implemented — HALT only.
- OS bootstrap (boot) not yet implemented; SCSI not modeled.

9.1.4 SRM Boot

Moniker: APP-DS10-SRMBOOT

```
[CON COM1 +    0.412 ms] probing hose 0, PCI
[CON COM1 + 8421.733 ms] Testing the System
```

```
[CON COM1 +    1.002 ms] initializing GCT/FRU at 3f32000
```

The bracketed timing prefix on each console line (for example, [CON COM1 + 8421.733 ms]) is the inter-line wall-clock delta on the COM1 console path. It makes the time-based firmware waits visible: small deltas are ordinary initialization steps, while large deltas mark the bounded device-wait and krn$_sleep loops described under Slow Boot and Firmware Hangs. The three lines above are illustrative; a full annotated power-up transcript is in Example - Performance.

9.1.5 SRM Commands >>> Prompt

Moniker: APP-DS10-SRMCMDS

```
>>>show (display the stored and active variable)
>>>set  (change the value of this variable)
```

Command	Usage / Value	Description
`auto_action`	• BOOT • HALT (TODO integration with : `$ touch /d/EmulatR/EmulatRAppUniV4/Emulatr/EMULATR_STOP)` • RESTART	Specifies the action the SRM console takes after a power-up, a system halt, or an unrecoverable error — i.e. whether the console stops at the `>>>` prompt or starts the operating system automatically. • **HALT** — Halts at the console prompt and waits for an operator command; the OS is not started automatically. This is the default. • **BOOT** — Automatically boots the operating system using *bootdef_dev* and *boot_osflags*. • **RESTART** — Attempts to restart a previously running operating system; if the restart fails, falls back to BOOT.

Command	Usage / Value	Description
		On EmulatR, HALT is the default and currently the only fully implemented action; BOOT and RESTART are planned.
boot_dev		Device, or device list, that the system will actually use for the next bootstrap. Normally derived by the console from *bootdef_dev*; you rarely set it directly.
boot_file		Name of the boot file (secondary bootstrap or kernel image) loaded during a bootstrap. Empty selects the operating system's default.
boot_osflags	0	Default operating-system boot flags passed to the OS at bootstrap (for OpenVMS, the system-root number and flag bits; for Tru64 UNIX, flags such as *a* for multi-user autoboot). Used when *boot* is issued with no *-flags* argument.
boot_reset	OFF	Whether a full hardware reset and re-initialization is performed before each bootstrap. **OFF** performs a warm boot; **ON** forces a cold reset first.
bootbios		Boots the alternate console / BIOS firmware image (AlphaBIOS / ARC) when one is present, instead of the SRM bootstrap. Firmware-specific.
bootdef_dev		Primary setting most operators use: the default boot device (or ordered device list) the console boots from when *boot* is typed with no device argument.

Command	Usage / Value	Description
booted_dev		Read-only. The device the system actually booted from during the last bootstrap.
booted_file		Read-only. The boot file actually loaded during the last bootstrap.
booted_osflags		Read-only. The boot flags actually in effect for the last bootstrap.
char_set	0	Console character-set selector. **0** selects the default (multinational) set.
com1_baud	9600	Line speed (baud rate) for serial port COM1.
com1_flow	NONE	Flow control for COM1: **NONE**, **SOFTWARE** (XON/XOFF), or **HARDWARE** (RTS/CTS).
com1_mode	SNOOP	Operating mode of COM1. **SNOOP** passively monitors the line in addition to normal console traffic.
com1_modem	OFF	Whether modem-control signals (DTR/DSR/DCD) are honored on COM1. **OFF** ignores modem signals.
com2_baud	9600	Line speed (baud rate) for serial port COM2.
com2_flow	SOFTWARE	Flow control for COM2: **NONE**, **SOFTWARE** (XON/XOFF), or **HARDWARE** (RTS/CTS).
com2_modem	OFF	Whether modem-control signals are honored on COM2. **OFF** ignores modem signals.
console	serial	Selects the console terminal: **serial** (a COM port) or **graphics** (attached VGA/keyboard). Determines where SRM console I/O is directed.
console_memory_allocation	old	Console heap-allocation scheme. **old** uses the legacy

Command	Usage / Value	Description
		memory layout; **new** uses the revised layout. Firmware-specific.
controlp	ON	Enables Ctrl/P as the console halt / attention character. **ON** lets Ctrl/P interrupt to the >>> prompt.
d_bell	off	Diagnostics: sound the console bell when a diagnostic error is detected.
d_cleanup	on	Diagnostics: perform resource cleanup (free buffers, reset devices) after a test run completes.
d_complete	off	Diagnostics: run each selected test to full completion rather than stopping at the first failure.
d_eop	off	Diagnostics: report at the end of each pass (end-of-pass messages).
d_group	field	Diagnostics: which test group to run, e.g. **field** (field-service set) or **mfg** (manufacturing set).
d_harderr	halt	Diagnostics: action taken on a hard (unrecoverable) error — **halt** stops, **continue** proceeds to the next test.
d_loghard	on	Diagnostics: record hard errors in the console error log.
d_logsoft	off	Diagnostics: record soft (recoverable) errors in the console error log.
d_omit		Diagnostics: list of individual tests to skip during a run.
d_oper	on	Diagnostics: allow operator-interactive tests (those that need a response or external loopback).

Command	Usage / Value	Description
d_passes	1	Diagnostics: number of times the selected test sequence is repeated.
d_quick	off	Diagnostics: run an abbreviated (quick) version of each test for faster coverage.
d_report	full	Diagnostics: report verbosity — **full**, **summary**, or **summary-only** output.
d_runtime	0	Diagnostics: maximum run time, in minutes, for the test sequence. **0** means no limit.
d_softerr	continue	Diagnostics: action taken on a soft error — **continue** or **halt**.
d_startup	off	Diagnostics: run the diagnostic sequence automatically at console startup.
d_status	off	Diagnostics: emit periodic in-progress status messages while tests run.
d_trace	off	Diagnostics: trace test execution (verbose per-step tracing) for debugging the diagnostics themselves.
d_verbose	0	Diagnostics: overall verbosity level; higher values produce more detailed output.
dump_dev		Device to which a system crash / memory dump is written.
enable_audit	ON	Enable the console audit / event log, recording significant console and boot events.
ffauto	OFF	Fast-fault automatic handling. When **ON**, the console automatically advances through the fail-fast / fault sequence without operator prompts. Firmware-specific.
ffnext	OFF	Fast-fault step control: advances to the next item in

Command	Usage / Value	Description
		the fail-fast sequence. Firmware-specific.
`full_powerup_diags`	ON	Run the complete power-up self-test suite. **ON** runs full diagnostics; **OFF** runs an abbreviated set for faster startup.
`heap_expand`	NONE	Policy for expanding the console heap when it runs low. **NONE** disables dynamic expansion. Firmware-specific.
`i`	g	User-defined console scratch variable (single-letter). Not interpreted by the console; available for operator scripts/macros.
`j`	w	User-defined console scratch variable (single-letter). Not interpreted by the console; available for operator scripts/macros.
`k`	g	User-defined console scratch variable (single-letter). Not interpreted by the console; available for operator scripts/macros.
`kbd_hardware_type`	PCXAL	Attached keyboard hardware type. **PCXAL** denotes a PS/2-style LK-layout keyboard.
`language`	36	Numeric console language code. The value selects the on-screen language used by the graphics console.
`language_name`	English (American)	Read-only. Human-readable name corresponding to the current *language* code.
`license`	MU	License class reported by the console. **MU** indicates a multi-user license.
`memory_test`	full	Depth of memory testing at power-up: **full**, **partial**, or

Command	Usage / Value	Description
		none.
`mfg_status`	19bcf8	Read-only. Manufacturing / test status word (hexadecimal), set during factory test.
`N1`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
`N10`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
`N11`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
`N12`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
`N13`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
`N14`		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.

Command	Usage / Value	Description
N15		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N16		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N2		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N3		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N4		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N5		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N6		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not

Command	Usage / Value	Description
		interpreted by the console itself.
N7		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N8		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
N9		General-purpose nonvolatile (NVRAM) console save slot. Reserved for firmware, PALcode, or operator use; not interpreted by the console itself.
oem_string		OEM identification string the console reports to software and in banners.
os_type	OpenVMS	Target operating system: **OpenVMS**, **UNIX** (Tru64), or **NT** (ARC/AlphaBIOS). Selects the PALcode variant and boot conventions the console uses.
page_table_levels	3	Number of virtual-memory page-table levels the console/PALcode maps. **3** is standard for Alpha with 8 KB pages.
pal	OpenVMS PALcode V1.98-83, Tru64 UNIX PALcode V1.92-73	Read-only. Version strings of the OpenVMS and Tru64 UNIX PALcode images currently loaded.
pci_parity	ON	Enable PCI-bus parity checking. **ON** reports PCI parity errors; **OFF** suppresses checking.

Command	Usage / Value	Description
prefetch_mode	ON	Enable hardware data prefetch. **ON** improves performance; **OFF** may be used for debugging.
prompt	>>>	The console prompt string. Default is >>>.
reset_boot_arg0		Saved boot argument 0, restored on system reset for an automatic re-boot.
reset_boot_arg1		Saved boot argument 1, restored on system reset for an automatic re-boot.
reset_boot_arg2		Saved boot argument 2, restored on system reset for an automatic re-boot.
rmc_halt	DISABLED	Remote Management Console halt control. **DISABLED** prevents the RMC from halting the CPU; **ENABLED** allows it.
scsi_poll	ON	Poll the SCSI bus for attached devices during console initialization.
scsi_reset	ON	Issue a SCSI bus reset during console initialization.
shutdown_temp	60	Temperature threshold, in degrees C, at which the system performs an automatic thermal shutdown.
srm2ctrl		Firmware mapping from an SRM device name to its controller. Used internally to translate console device paths.
srm2dev		Firmware mapping from an SRM device name to its underlying device. Used internally to translate console device paths.
sys_serial_num	test123	System serial number string stored in NVRAM.
tt_allow_login	1	Allow interactive login on the console terminal. **1** permits

Command	Usage / Value	Description
		login; **0** denies it.
`tty_dev`	0	Active console terminal device number. **0** selects COM1; **1** selects COM2.
`user_def1`		Free-form user-defined variable for site-specific use; not interpreted by the console.
`user_def2`		Free-form user-defined variable for site-specific use; not interpreted by the console.
`version`	V7.3-2 Feb 27 2007 13:25:53	Read-only. SRM console firmware version and build date.
`wwid0`		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
`wwid1`		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
`wwid2`		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
`wwid3`		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
`wwid4`		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the

Command	Usage / Value	Description
		same target device across reboots.
wwid5		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
wwid6		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.
wwid7		Stores a persistent device Worldwide ID (WWID) for a Fibre Channel / SAS boot path, so the console can locate the same target device across reboots.

9.1.6 SuperIO Controller

Moniker: APP-DS10-SUPERIO

#22 SuperIO — word-fold CONFIRMED.

indicated by the w= column:

```
w=2 val=0x22    -> index 0x22 (Power Control), data 0x00  ← Table 61
default ✓
w=2 val=0x23    -> index 0x23 (Power Mgmt),    data 0x00  ← default ✓
w=2 val=0x342D  -> index 0x2D (TEST1),         data 0x34
```

The SuperIO is configured byte ordered 100% locked. Reading as (`index=low, data=high`):

```
w=2 0x07   -> LDN=0     (select FDD)
w=2 0x360  -> reg0x60=0x03   ┐ base 0x03F0  ← the 0x60/0x03 pair, byte
order confirmed
w=2 0xF061 -> reg0x61=0xF0   ┘
w=2 0x670  -> reg0x70=0x06   IRQ6
w=2 0x274  -> reg0x74=0x02   DMA2
w=2 0x130  -> reg0x30=0x01   Activate (FDD enabled!)
w=2 0x307  -> LDN=3 ... 0xBC61 -> base 0x03BC (LPT1), IRQ7
w=2 0x407  -> LDN=4 ... 0xF861 -> base 0x03F8 (COM1), IRQ4
```

```
w=2 0x507  -> LDN=5 ... 0xF861 -> base 0x02F8 (COM2), IRQ3
```

Considerations:

It's programming every logical device —

FDD@3F0/IRQ6/DMA2	LPT@3BC, COM1@3F8	COM2@2F8

— exactly matching show config. The fold is confirmed and the byte order is low=index, high=data beyond doubt.

Trace Evidence:

0 → 93 SIO-TRACE lines, every one [cfg], the word-fold split cleanly into index→0x3F0 / data→0x3F1. The FDD logical device configures at 0x03F0/IRQ6/DMA2, detect matches the seeded device-ID, and the suite stayed at 407. The SuperIO is properly modeled — is grounded end-to-end in the HRM trace.

```
SIO-TRACE W port=0x3F0 val=0x60 [cfg]   ← index 0x60
SIO-TRACE W port=0x3F1 val=0x03 [cfg]   ← data 0x03   = FDD base high
SIO-TRACE W port=0x3F0 val=0x61 [cfg]
SIO-TRACE W port=0x3F1 val=0xF0 [cfg]   = base 0x03F0
SIO-TRACE W port=0x3F0 val=0x70 [cfg]
SIO-TRACE W port=0x3F1 val=0x06 [cfg]   = IRQ6
```

The config registers are 16-bit writes, low byte = index, high byte = data — exactly the fold, and the byte order is locked (the 0x22/0x23 data bytes match their Table 61 defaults). The byte-split is the right fix.

9.1.7 Task Matrix

Moniker: APP-DS10-TASKMATRIX

1. dqa0 enumeration gap — trace 0x1F0–0x1F7 + PCI config to bus0/dev5/func1; classify port-never-init vs probe-protocol.
2. Fix dqa0 + absent-slave probe — per trace result; float absent-slave scratch regs so the 0xAA presence gate fails fast (blocked by #1).
3. Verify — suite green + dqa0 visible in show device + snapshot written (blocked by #2, #3).
4. 4.Boot-phase timeout loops — enumerate/instrument (GCT/FRU, dq waits, IIC, SCSI, RSCC balloon); mostly collapses once #2/#5 land.
5. SRM update non-persistence — the LFU TX→RX loopback (prompt bytes read as input) + confirm dtor-flush path.
6. Gate output behind env var — silent by default; kill the per-byte fprintf+fflush console mirror.
7. Profile cold-boot→>>> — best/worst case, --no-autoload, output gated; establish IPS baseline.

8. Scope 10x IPS — Tier-1 (gating/build/dispatch) then the V5 two-tier JIT question (blocked by #10, #8).
9. >>> halt clean shutdown — verify CALL_PAL HALT exits through ~Machine (flush) cleanly (no sentinel touch needed).
10. Flash ROM writes are disabled Needs to be investigated
11. dqa0.0.0.105.0 has no media present or is disabled via the run/stop switch - after wards we have a stall - timer-loop?
12. initializing GCT/FRU at 3f3200 is significantly delayed.
13. #21 (still open): the Checking dva0... for option firmware files line is followed by a krn$_sleep-based option-firmware scan wait — the 0x7bef0 software-tick loop. That's a time-based sleep, not a device probe, so no amount of FDC modeling touches it. Your own profile pins it: the gap after that line was +2158903 ms ≈ 36 min.

9.1.8 TimeOut Loops

Moniker: APP-DS10-TIMEOUT

The timeout loops have concrete bounds.

Loop Function	Approx. Wall Clock	Description
dq_wait_on_busy	5000×100µs	
dq_wait_for_drq	5000×100–200µs	
dq_long_wait_on_busy	310000×100µs ≈ 31s	

9.2 Appendix DS20

Moniker: APP-DS20

Planned. DS20 is not yet modeled or validated in EmulatR. [TODO: add platform manifest and reference notes once available.]

9.3 Appendix ES40

Moniker: APP-ES40

Planned. Platform manifest es40_platform.json is scaffolded; the platform has not yet been driven to >>>. [TODO: validate and document.]

9.4 Appendix ES45

Moniker: APP-ES45

Planned. Platform manifest es45_platform.json (EV67) is scaffolded; not yet validated. [TODO: validate and document.

Index

- 1 -

- 2 -

- A -

- B -

- C -

- D -

- E -

- F -

- G -

- H -

- I -

- J -

- K -

- L -

- M -

- N -

- O -

- P -

- Q -

- R -